

2024

PHYSICS — HONOURS — PRACTICAL

Paper : SEC-1P

(Introduction to Graph Plotting and Programming)

Full Marks : 100

The figures in the margin indicate full marks.

DAY - 3

LNB - 20, VIVA - 20, Programming - 60

Answer any one question from each group.

Group - A

1. Make the following data table in a file.

x	-4.8	-3.4	-2.5	1.0	2.5	3.1	3.8	4.7
y	60.0	29.5	14.0	6.2	24.5	36.0	52.1	76

Plot the data in second column against that of the first column. Set the range of x values from -5 to $+5$. Try to fit the data with the equation of type $f(x) = ax^2 + bx + c$. Show the points and the fitted line on the same plot. Mention the values of a , b and c after fitting the data. Store the plot in a file and print it.

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2. Define the functions $f1(x) = \sin\sqrt{2|x|}$, $f2(x) = e^{-\frac{x}{10}} \cos 3x$ and $f3(x) = x \log_{10}|x|$. Set x -range from -5 to 5 and y -range from -5 to 5 . Set samples 1000. Show both the axes on the plot. Ticks of the x -axis and y -axis will be at interval of 0.5 and 1.0 and placed on the axes.

Write a GNUPLOT script to plot the function and save the plot in a file.

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3. Plot the following polar function for $0 \leq \theta \leq 2\pi$ with sample 800.

$$\rho(\theta) = \sin 2\theta \cos 2\theta$$

Put the title, x -label and y -label as 'Polar plots', ' θ ' and ' $r(\theta)$ ' respectively. Set sample 800. Save the plot in suitable format.

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4. Plot the following piecewise discontinuous function for $-2 \leq x \leq 4$ with sample 800.

$$\begin{aligned} f(x) &= x^2 - 1 \text{ for } x < 1 \\ &= x^3 - 5 \text{ for } 1 < x < 2 \\ &= 5 - 2x \text{ for } x > 2 \end{aligned}$$

Set the title 'Piecewise discontinuous functions'. Set x-label, y-label as 'x', and 'f(x)' respectively.

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5. Plot the following parametric functions with $0 \leq t \leq 2\pi$

$$x(t) = \sin 2t \cos t, \quad y(t) = \sin 2t \sin t$$

Put the title as 'Parametric plot'. Set samples to 800. Save the plot in suitable format.

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6. Save the following data table in a file :

x	-1.0	-0.7	-0.4	-0.1	0.2	0.5	0.8	1.1	1.4	1.7	2.0
f(x)	0.184	0.243	0.306	0.369	0.426	0.47	0.495	0.499	0.48	0.442	0.389

Fit the data points of the second row against that of the first row using GNUPLOT with the equation

$$f(x) = \frac{1}{a} e^{-\frac{(x-b)^2}{a^2}}. \text{ Write the values of the fitted parameters } a, b. \text{ Show the fitted curve along with the}$$

data points in the same plot. Set the title, x-label, y-label as 'Data fitting', 'x' and 'f(x)' respectively. Save the plot in suitable format.

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7. Define the following function
$$f(x) = \begin{cases} x \cos x & x < 0 \\ x - 1 + \cos x & 0 \leq x \leq \frac{\pi}{2} \\ \sin x + x + \frac{\pi}{2} & x > \frac{\pi}{2} \end{cases}$$

Set the range of x values from $-\pi$ to π and the range of y values from $-\pi$ to $+2\pi$. Show both the axes on the plot. Set title 'Continuous function without derivative at all points'. Plot the function, store it in a file and print it.

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Group - B

8. (a) Using Newton-Raphson method with starting point $x_0 = 0$, determine the root of the following equation correct up to three decimal places :

$$\ln(x^2 + 2) - e^{-\frac{x}{2}} = 0 \quad 30$$

Print the root.

- (b) Write a Python program to calculate the value of $e^{2.31}$ using the following infinite series

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

correct up to three decimal places. Print the value.

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9. (a) Write a Python program to find the root of the transcendental equation correct up to three decimal places

$$x^2 \sin 2x - \cos 2x = 0$$

in between $[-2, 0]$ using Bisection method. Print the root.

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- (b) Multiply the following matrices.

$$A = \begin{pmatrix} -1 & 4 \\ 3 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

Print the multiplied matrix.

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10. (a) A first order differential equation is given as below :

$$\frac{dy}{dx} = x + y \quad \text{with } y(0) = 0$$

Create a list (say X) with 100 x -values between $[0,1]$. Write a Python program to solve this differential equation using Euler method for x -values in the list X . It is given that the exact solution is $y(x) = e^x - x - 1$. Compare the calculated and exact values at $x = 1$.

Print both the values.

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- (b) Calculate the highest prime number within the range $[20, 80]$. Print this number.

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11. (a) Using Newton-Raphson method with starting point $x_0 = -4$, determine the root of the following equation correct up to three decimal places :

$$5x \sin x - 3x = 0$$

Print the root.

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- (b) Arrange the following numbers in descending order.

4.82, - 7.53, 8.39, 2.93, - 1.56, - 5.62

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12. (a) A first order differential equation is given as below :

$$\frac{dy}{dx} = xe^{-y} \quad \text{with } y(0) = 0$$

Create a list (say X) with 100 x -values between $[0,1]$. Write a Python program to solve this differential equation using Euler method for x -values in the list X . It is given that the exact solution is $y(x) = \ln(x^2/2 + 1)$. Compare the calculated and exact values at $x = 1$.

Print both the values.

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- (b) Determine the number of the letter 't' in the following sentence 'Python is great'. Print that number.

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13. (a) Write a python program to find the root of the transcendental equation correct up to three decimal places

$$x^3 - 2x^2 + 5x - 7 = 0$$

in between (1, 2) using Bisection method. Print the root.

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- (b) Write a Python program to determine the factors of the integer 48. Print all the factors.

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14. (a) Write a Python program to calculate the matrix B satisfying the following relation

$$B = 3A^2 + 2A$$

for

$$A = \begin{pmatrix} -2 & 1 & 4 \\ 3 & 0 & -2 \\ 4 & -2 & -1 \end{pmatrix}$$

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Print the matrix B .

- (b) Create a list (A) with the following numbers :

$$A = \left[-\frac{\pi}{2}, 2\pi, 0, \frac{3\pi}{2}, -\pi, \frac{\pi}{2} \right]$$

Create another list (B) of $\sin x_i$, where x_i are the elements of list A . Print list B .

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